CSE 200 : Online - 1 on LaTeX (A1)

Time: **50 Minutes**
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Mark Distribution

Topic	**Marks**
Document Structure, Calligraphic Title & Two-column Layout	5
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Equations, Cases, Matrices & Mathematical Notation	14
Multirow and Multicolumn Table	6
Rotated Figure, Caption & Label	4
Algorithm and Complexity Expression	5
Python Code Listing and Syntax Formatting	6
Citations & Bibliography	2
Cross-references	2
Overall Presentation & Successful Compilation	2
Total	**50**

Knot Theory

Diagrams, Invariants, and Applications

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1 Knots as Mathematical Objects

1.1 From Rope to Curve

A mathematical knot is a closed curve in three-dimensional space that has no self-intersections. Formally, it may be represented by a continuous one-to-one map

$$\mathscr{K} : S^1 \longrightarrow \mathbb{R}^3, \tag{1}$$

where S^1 denotes the unit circle. Two knots are considered equivalent if one can be continuously deformed into the other without cutting the curve or allowing it to pass through itself [1].

The simplest knot is the *unknot*, which can be deformed into an ordinary circle. A nontrivial example is the trefoil knot shown in Figure 1.

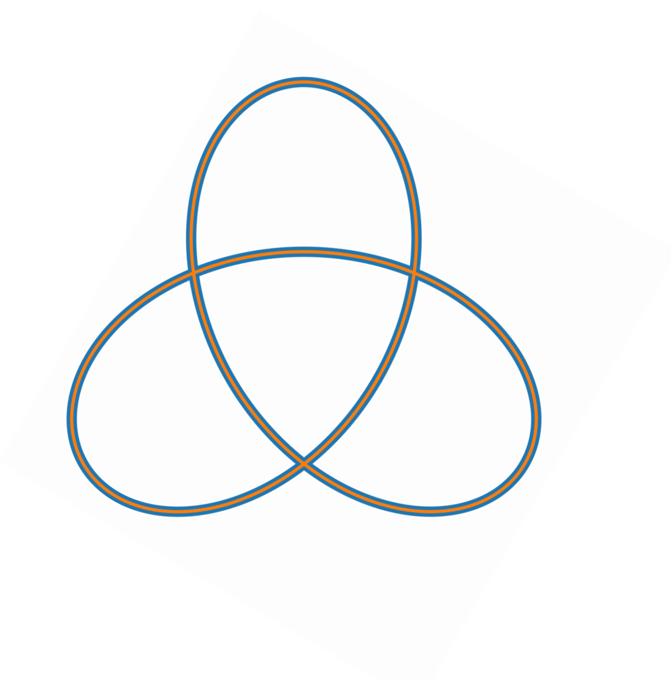

Figure 1: A planar drawing of a trefoil knot.

1.2 Diagrams and Crossings

A knot diagram is a projection of a spatial knot onto a plane. At every crossing, a small break indicates which strand passes underneath. The main diagrammatic objects are:

`Arc`	A continuous segment between two undercrossings.
`Crossing`	A point where one projected strand passes over another.
`Orientation`	A selected direction of travel around the knot.

A diagram may contain many crossings even when the underlying knot is simple, while an invariant must remain unchanged under every legal deformation of the knot.

2 Crossing Number and Writhe

2.1 Crossing Number

For a diagram D, let $c(D)$ be its number of crossings. The crossing number of a knot K is

$$c(K) = \min_{D \sim K} c(D), \tag{2}$$

where $D \sim K$ means that D represents the knot K. Thus,

$$c(\text{unknot}) = 0, \qquad c(\text{trefoil}) = 3.$$

2.2 Signed Crossings

For an oriented diagram, each crossing is assigned a sign $\varepsilon_i \in \{+1, -1\}$. The writhe of the diagram is

$$w(D) = \sum_{i=1}^{c(D)} \varepsilon_i = n_+(D) - n_-(D), \tag{3}$$

where $n_+(D)$ and $n_-(D)$ count positive and negative crossings. For example, if the signs are $+1, +1, -1, +1$, then

$$w(D) = 1 + 1 - 1 + 1 = 2.$$

Table 1: Selected knots and several elementary invariants.

Family	Knot	Selected invariants			Chiral?
		$c(K)$	$\det(K)$	$\Delta_K(t)$	
Basic	Unknot	0	1	1	No
Torus knots	Trefoil 3_1	3	3	$t^{-1}-1+t$	Yes
	Cinquefoil 5_1	5	5	$t^{-2}-t^{-1}+1-t+t^2$	Yes
Twist knot	Figure-eight 4_1	4	5	$-t^{-1}+3-t$	No

Writhe is useful in calculations, but it is not itself a knot invariant: it can change under a Reidemeister move of type I.

3 Reidemeister Moves

Two diagrams represent equivalent knots exactly when one can be changed into the other by planar deformation together with the three Reidemeister moves [2]:

(I) **Twist move:** adds or removes one loop and one crossing;

(II) **Poke move:** adds or removes two neighboring crossings;

(III) **Slide move:** moves one strand across a crossing.

These moves preserve the underlying knot type even though the visible number and arrangement of crossings may change.

4 Knot Invariants

4.1 Alexander Polynomial

A knot invariant assigns the same value to every diagram of the same knot. One important example is the Alexander polynomial $\Delta_K(t)$, defined up to multiplication by $\pm t^m$. A common normalization satisfies

$$\Delta_K(t) = \Delta_K(t^{-1}), \qquad \Delta_K(1) = 1. \tag{4}$$

For the trefoil knot,

$$\Delta_{3_1}(t) = t^{-1} - 1 + t. \tag{5}$$

The determinant of a knot is obtained from the polynomial by

$$\det(K) = |\Delta_K(-1)|. \tag{6}$$

Hence,

$$\det(3_1) = |-1-1-1| = 3.$$

Table 1 compares these quantities for several small knots.

4.2 A Skein Relation

Many polynomial invariants are computed by comparing three diagrams that differ only inside a small region. For the Alexander polynomial, one form of the skein relation is

$$\Delta_{L_+}(t) - \Delta_{L_-}(t) = \left(t^{1/2} - t^{-1/2}\right)\Delta_{L_0}(t), \tag{7}$$

where L_+, L_-, and L_0 denote a positive crossing, a negative crossing, and a smoothing, respectively.

4.3 The Signature of a Knot

Another useful knot invariant is the signature $\sigma(K)$, obtained from a Seifert matrix V. It is defined as the signature of the symmetric matrix

$$V + V^T,$$

that is, the number of positive eigenvalues minus the number of negative eigenvalues.

For a knot and its mirror image, the signature satisfies

$$\sigma(K) = \begin{cases} 0, & \text{if } K \text{ is the unknot,} \\ -2, & \text{if } K \text{ is the right-handed trefoil,} \\ 2, & \text{if } K \text{ is the left-handed trefoil.} \end{cases} \tag{8}$$

In particular, the signature distinguishes the right-handed trefoil from its mirror image, even though the two knots have the same Alexander polynomial.

5 A Small Diagram Algorithm

Algorithm 1 computes the writhe when the signed crossings of an oriented diagram are already known.

Algorithm 1 Compute the writhe of a knot diagram

Require: A list $S = (\varepsilon_1, \ldots, \varepsilon_m)$ with $\varepsilon_i \in \{-1, +1\}$
Ensure: The writhe $w(D)$

```
w ← 0
for each ε in S do
    w ← w + ε
end for
return w
```

The algorithm performs one addition per crossing, so its running time is

$$T(m) = \Theta(m), \tag{9}$$

where $m = c(D)$.

6 A Short Python Implementation

Listing 1 implements the same calculation. It also checks that every crossing sign is valid.

```python
def writhe(crossing_signs):
    """Return the writhe of an oriented knot diagram."""
    if any(
        sign not in (-1, 1) for sign in crossing_signs
    ):
        raise ValueError("Each sign must be +1 or -1")
    return sum(crossing_signs)


if __name__ == "__main__":
    signs = [1, 1, -1, 1]
    print(writhe(signs)) # Output: 2
```

Listing 1: Computing the writhe from signed crossings.

The mathematical definition in Equation (3) and the code in Listing 1 perform the same summation.

7 Applications Beyond Pure Mathematics

Knot theory appears whenever long flexible objects become entangled. Examples include:

- circular DNA molecules and the action of topoisomerase enzymes;
- polymer chains and molecular entanglement;
- vortex lines in fluids and plasmas;
- knot-based models in quantum computation.

For two oriented components C_1 and C_2 in a link diagram, the linking number can be calculated from crossings between the two components:

$$\mathrm{lk}(C_1, C_2) = \frac{1}{2} \sum_{p \in C_1 \cap C_2} \varepsilon(p). \tag{10}$$

This quantity measures how many times the two components wind around one another, with orientation taken into account.

For a link $L = C_1 \cup \cdots \cup C_n$, the pairwise linking numbers may be collected into the symmetric linking matrix

$$\Lambda(L) = \left[\lambda_{ij}\right]_{i,j=1}^{n}, \qquad \lambda_{ij} = \begin{cases} \mathrm{lk}(C_i, C_j), & i \neq j, \\ 0, & i = j. \end{cases} \tag{11}$$

References

[1] C. C. Adams, *The Knot Book: An Elementary Introduction to the Mathematical Theory of Knots*, 2nd ed. American Mathematical Society, 2004.

[2] L. H. Kauffman, *On Knots*. Princeton University Press, 1987.